

Assignment 5a

Sunday, 31 March 2024 3:42 pm

Assignment 5a

Refer to Lecture 20, page 6 onwards for this

Q1, Q2, (not Q3), Q4, Q5, (not Q6), Q7

Q. no. 1 (a)

CHECK WHETHER $\underline{V_1} = (1, 1, 2)$,
 $\underline{V_2} = (1, 0, 1)$, AND $\underline{V_3} = (2, 1, 3)$
SPAN THE VECTOR SPACE $\mathbb{R}^3$

No, since $V_3 = V_1 + V_2$

Q. no. 1 (b)

ARE THE FOLLOWING TRUE OR FALSE?

(I) A SET THAT CONTAINS THE ZERO VECTOR IS LINEARLY DEPENDENT.

True, $k_1 \underline{V_1} + \dots + k_n \underline{V_n} + c_1 \underline{0} = \underline{0} \quad \because \forall k_i = 0$
but $c_1 \neq 0$

(II) IF $\underline{W}$ IS A SUBSPACE OF V , THEN $\dim(W) \leq \dim(V)$;
MOREOVER, $\dim(W) = \dim(V)$;
IF AND ONLY IF $W = V$

IF AND ONLY IF $W=U$
True, Number of bases are equal.

(III) EVERY NONZERO FINITE DIM.
ENSIONAL INNER PRODUCT SPACE
HAS AN ORTHONORMAL BASIS.

True, by Gram Schmidt Process

(IV) THE PRODUCT $A\underline{x}$ IN A LINEAR
SYSTEM IS A LINEAR COMBINA-
TION OF THE COLUMN VECTORS
OF A .

True, $A = [C_1 | C_2 | \dots | C_n] \Rightarrow Ax = [C_1x | \dots | C_nx]$

(V) A SYSTEM OF LINEAR EQUA-
TIONS $A\underline{x} = \underline{b}$ IS CONSISTENT
IF AND ONLY IF $\underline{b}$ IS IN THE
COLUMN SPACE OF A .

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True, $C_1x = b_1, C_2x = b_2, \dots, C_nx = b_n$

(VI) $A\underline{x} = \underline{b}$ IS CONSISTENT
IF AND ONLY IF THE RANK
OF THE COEFFICIENT MATRIX
 A IS THE SAME AS THE
RANK OF THE AUGMENTED
MATRIX $[A/\underline{b}]$.

True, If Rank increase then math error
[Q.no.2]

Q.no.2

LET $A = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$

FIND:

- (a) ECHELON FORM OF A .
 (b) BASIS FOR THE COLUMN SPACE OF A BY WATCHING THE

COLUMN VECTORS IN A WHICH CORRESPOND TO THE COLUMN VECTORS IN ECHELON FORM (CONTAINING THE LEADING 1's). (8)

c) CAN WE ALSO FIND THE BASIS FOR THE ROW SPACE OF A (CONSISTING ENTIRELY OF ROW VECTORS FROM A) BY THE METHOD (USED IN (b))?
 EXPLAIN YOUR ANSWER?

Question 02

Solution: Echelon form of a matrix

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Column space of a matrix is $\{\{1, 2, 0, 2\}, \{-2, -5, 5, 6\}, \{0, 3, 15, 18\}\}$

Row space of $\{\{1, -2, 0, 0, 3\}, \{0, 1, 3, 2, 0\}, \{0, 0, 1, 1, 0\}\}$

Q.no.4 (INNER PRODUCT SPACE)

(a) SEE THE DEFINITION OF WEIGHTED EUCLIDEAN INNER PRODUCT.
 (P. 277 8th ED. OR P. 298 7th ED.)

(b) LET $\underline{u} = (u_1, u_2)$ AND $\underline{v} = (v_1, v_2)$ BE VECTORS IN $\mathbb{R}^2$. VERIFY THAT THE WEIGHTED EUCLIDEAN INNER

PRODUCT

$\langle \underline{u}, \underline{v} \rangle = 3u_1v_1 + 2u_2v_2$
 SATISFIES THE FOUR INNER PRODUCT AXIOMS.

Solution: $\langle u, v \rangle = 3u_1v_1 + 2u_2v_2 = 3v_1u_1 + 2v_2u_2 = \langle v, u \rangle$

$\langle (u+v), w \rangle = 3(u_1+v_1)w_1 + 2(u_2+v_2)w_2 = 3u_1w_1 + 2u_2w_2 + 3v_1w_1 + 2v_2w_2 =$

$\langle u, w \rangle + \langle v, w \rangle$

$\langle ku, v \rangle = 3ku_1v_1 + 2ku_2v_2 = k(3u_1v_1 + 2u_2v_2) = k\langle u, v \rangle$

$\langle u, u \rangle = 3u_1^2 + 2u_2^2 \geq 0$

$\langle ku, v \rangle$

$3(ku_1)v_1 + 2(ku_2)v_2$

$k(3u_1v_1 + 2u_2v_2)$
 $k\langle u, v \rangle$

Q.no.5

(a) IF $U = \begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}$ AND $V = \begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$

ARE 2×2 MATRICES, THEN PROVE THAT THE FOLLOWING FORMULA DEFINES AN INNER PRODUCT ON $M_{22} \rightarrow (\underline{u} = U, \underline{v} = V)$

$\langle \underline{u}, \underline{v} \rangle = u_1v_1 + u_2v_2 + u_3v_3 + u_4v_4$

Similar to previous answer, verify all four

Similar to previous answer, verify all four properties.

(b) If $P = a_0 + a_1x + a_2x^2$ AND $Q = b_0 + b_1x + b_2x^2$ ARE ANY TWO VECTORS IN P_2 , THEN PROVE THAT THE FOLLOWING FORMULA DEFINES AN INNER PRODUCT ON P_2

$$\langle P, Q \rangle = a_0b_0 + a_1b_1 + a_2b_2$$

Same as above, verify all four properties of inner product space

Q. no. 7

FOR ANY INNER PRODUCT SPACE

PROVE THAT

$$(a) \quad \|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$$

$$(u+v) \cdot (u+v) = \|u\|^2 + 2u \cdot v + \|v\|^2$$

$$(u-v) \cdot (u-v) = \|u\|^2 - 2u \cdot v + \|v\|^2$$

$$(b) \quad \langle u, v \rangle = \frac{1}{4} \|u+v\|^2 - \frac{1}{4} \|u-v\|^2 \quad \langle u, v \rangle$$

Question (a)

$$\text{Solution: } \|u+v\| + \|u-v\| = [(u+v) \cdot (u+v)]^{1/2} + [(u-v) \cdot (u-v)]^{1/2} = 2\|u\|^2 + 2\|v\|^2$$

(b)

$$\|u+v\|^2 - \|u-v\|^2$$

$$(u+v)(u+v) - (u-v)(u-v)$$

$$u(u+v) + v(u+v) - u(u-v) + v(u-v)$$

$$u^2 + uv + uv + v^2 - u^2 + uv + uv - v^2$$

$$4(u \cdot v)$$

$$\|u+v\|^2 - \|u-v\|^2 \text{ proved!}$$

$$4 \quad (\underline{u} \cdot \underline{v}) \\ \underline{u} \cdot \underline{v} = \frac{1}{4} (\|\underline{u} + \underline{v}\|^2 - \|\underline{u} - \underline{v}\|^2) \text{ proved!}$$